

Sales Forecasting & Demand Intelligence Report

Prepared for: Head of Supply Chain and CFO

Executive summary

Sales are growing, but demand is highly seasonal: the business consistently sells much more in the final quarter, especially November and December, and much less in January and February. Our forecast expects a normal post-holiday slowdown in January and February 2019, followed by a recovery in March. The best forecasting model estimates sales of about \$46,782, \$40,285, and \$72,234 for the next three months. For Supply Chain, this means inventory should be built earlier for seasonal peaks and managed more tightly after the holidays. For Finance, the main opportunity is to reduce working capital tied up in low-demand products while protecting availability for the highest-volume and fastest-moving categories.

Key findings from sales exploration and forecasting

- Technology is the largest revenue category at \$827,456, ahead of Furniture (\$728,659) and Office Supplies (\$705,422).
- East is the most consistent growth region, with average yearly growth of 18.1% and low year-to-year variation of 1.8%. West grows faster at 21.4%, but is much more volatile.
- Seasonality is strong. Average monthly sales are highest in November (\$87,540), December (\$80,370), and September (\$75,026); February is the weakest month at \$14,843.
- Forecast model selected: SARIMA. It had the best overall error balance, with MAE of \$14,862 and RMSE of \$17,300 on the recent test period. In plain terms, monthly forecasts are usually off by about \$15K to \$17K, so they should guide planning rather than replace manager judgment.
- Fulfillment timing is stable. Average order-to-ship time is about 4.0 days, with only small differences across regions. The bigger operational issue is demand timing, not shipping speed.

3-month sales forecast with confidence ranges

Month	Expected sales	Likely low	Likely high	Business meaning
2019-01	\$46,782	\$16,993	\$76,572	Post-holiday slowdown; control replenishment
2019-02	\$40,285	\$9,596	\$70,975	Weakest seasonal period; avoid overbuying
2019-03	\$72,234	\$41,319	\$103,150	Recovery month; prepare replenishment early

Plain-language interpretation: the middle estimate is the planning number. The low and high values show a reasonable range the business should be prepared for. Because the historical monthly dataset is small, the ranges are intentionally wide and should be combined with known promotions, vendor constraints, and management plans.

Top 3 anomalies detected and likely causes

- **2015-03-22:** weekly sales of \$37,704. Likely cause: early-year promotional or large order activity compared with the low starting baseline.
- **2018-12-02:** weekly sales of \$35,999. Likely cause: holiday shopping and year-end purchasing continued from the November peak.
- **2018-11-18:** weekly sales of \$30,572. Likely cause: Black Friday, Cyber Monday, and seasonal promotions likely drove unusually high demand.

Product demand segmentation and stocking strategy

Products were grouped by sales volume, volatility, order value, and growth. This turns the product list into practical stocking rules instead of treating every sub-category the same.

Demand segment	What it means	Recommended stocking strategy
High Volume, Stable Demand	Reliable sellers such as Accessories, Binders, Chairs, Phones, Storage, and Tables.	Keep strong availability. Use continuous replenishment and maintain at least two weeks of safety stock.
Volatile / Bursty Demand	High-value items with sharp demand swings, including Copiers and Machines.	Do not overstock all year. Buy to forecast, monitor weekly, and add extra coverage before Q4 or promotions.
Growing Demand	Sub-categories with strong recent growth, especially Supplies.	Increase base stock gradually, review monthly, and secure supplier capacity before demand accelerates.
Low Volume, Steady	Long-tail items such as Art, Envelopes, Fasteners, Labels, Paper, Furnishings, Appliances, and Bookcases.	Limit inventory investment. Use smaller reorder quantities, make-to-order, or drop-ship where possible.

Three concrete business recommendations

- **Build Q4 inventory earlier, not reactively.** November averages \$87,540 and December averages \$80,370, compared with only \$14,843 in February. Supply Chain should lock vendor capacity and inbound plans by August for seasonal peak items.
- **Prioritize Technology for growth capital and availability.** Technology produced \$827,456, the highest category revenue. CFO should protect budget for this category, while Supply Chain should reduce stockout risk on fast-moving Technology sub-categories such as Phones and Accessories.
- **Use segment-specific inventory rules instead of one blanket safety-stock target.** High-volume stable products need high service levels, while low-volume steady products should carry leaner stock. This approach frees working capital without putting core revenue items at risk.

Business risk / limitation

The system is based on historical sales from 2015 to 2018 and only 48 monthly data points. It cannot automatically know about future events such as new promotions, supplier disruptions, competitor actions, pricing changes, or economic shocks. For this reason, forecasts should be used as a planning baseline and reviewed by business leaders before final purchasing or budget decisions.

How leaders should use this report

Use the forecast to set the monthly purchasing baseline, use the anomaly report to investigate unusual demand events, and use the demand segments to set different stocking policies by product group. The best results will come from combining this system with known business events such as promotions, vendor lead times, and finance targets.